

Hourly Power Demand Prediction in Greece: Custom C/CUDA Neural Network Benchmark Report

Overview

This report evaluates the performance of a custom C/CUDA neural network model using real-world hourly system load logs. The dataset comprises sequential weekly records covering the Greek transmission system, **spanning from Saturday, August 15, 2026, through Friday, August 21, 2026**. Each entry captures hourly operational data, tracking actual system load versus model predictions, absolute deviations, and relative error percentages across varying conditions.

Data Source: System load logs provided by AΔMHE

1. <https://www.admie.gr/agora/statistika-agoras/file-download-api>
2. <https://www.admie.gr/getOperationMarketFilewRange?dateStart=2025-07-19&dateEnd=2026-08-22&FileCategory=RealTimeSCADASystemLoad>

Performance Logs by Day

The following tables present a detailed hourly breakdown comparing actual system load against model predictions for each day of the examined period:

Actual Load (MW)	Prediction (MW)	Error (MW)	Percent (%)	Time
Friday 21/8/2026				
5699	5714	-15	0,26320407	1:00
5413	5524	-111	2,05061888	2:00
5201	5355	-154	2,96096904	3:00
5054	5194	-140	2,7700831	4:00
4991	5043	-52	1,04187538	5:00
5044	4929	115	2,27993656	6:00
4955	4893	62	1,25126135	7:00
4421	4734	-313	7,07984619	8:00
3723	4163	-440	11,818426	9:00
3144	3423	-279	8,8740458	10:00
2834	2854	-20	0,7057163	11:00
2849	2617	232	8,14320814	12:00
3103	2737	366	11,7950371	13:00
3394	3149	245	7,2186211	14:00
3769	3681	88	2,33483683	15:00
4513	4346	167	3,70042101	16:00
5461	5326	135	2,47207471	17:00
6472	6314	158	2,44128554	18:00
7144	7089	55	0,76987682	19:00
7371	7436	-65	0,88183422	20:00
7302	7431	-129	1,76663928	21:00
6939	7223	-284	4,09280876	22:00
6576	6855	-279	4,24270073	23:00
Thursday 20/8/2026				
5992	6494	-502	8,37783712	0:00
5596	5656	-60	1,07219442	1:00
5317	5465	-148	2,78352454	2:00
5123	5303	-180	3,51356627	3:00
4994	5159	-165	3,30396476	4:00
4970	5027	-57	1,14688129	5:00
5084	4950	134	2,63571991	6:00
5096	4971	125	2,45290424	7:00
4626	4909	-283	6,1175962	8:00
3949	4420	-471	11,9270701	9:00
3316	3708	-392	11,8214717	10:00

2948	3072	-124	4,20624152	11:00
2946	2769	177	6,00814664	12:00
3125	2873	252	8,064	13:00
3324	3221	103	3,09867629	14:00
3635	3669	-34	0,93535076	15:00
4206	4273	-67	1,59296243	16:00
5180	5103	77	1,48648649	17:00
6452	6140	312	4,83570986	18:00
7294	7117	177	2,42665204	19:00
7542	7554	-12	0,15910899	20:00
7354	7545	-191	2,597226	21:00
6925	7265	-340	4,90974729	22:00
6521	6850	-329	5,04523846	23:00
Wednesday 19/8/2026				
5901	6458	-557	9,43907812	0:00
5561	5591	-30	0,53947132	1:00
5278	5436	-158	2,99355817	2:00
5116	5277	-161	3,14698984	3:00
5015	5162	-147	2,93120638	4:00
4978	5059	-81	1,6271595	5:00
5076	4980	96	1,89125296	6:00
5099	4995	104	2,03961561	7:00
4671	4950	-279	5,97302505	8:00
4025	4511	-486	12,0745342	9:00
3411	3837	-426	12,4890062	10:00
3012	3214	-202	6,7065073	11:00
2945	2872	73	2,47877759	12:00
3074	2916	158	5,13988289	13:00
3205	3218	-13	0,40561622	14:00
3503	3599	-96	2,74050814	15:00
4033	4189	-156	3,86808827	16:00
5040	4985	55	1,09126984	17:00
6272	6065	207	3,30038265	18:00
7133	7032	101	1,41595402	19:00
7442	7491	-49	0,65842515	20:00
7315	7514	-199	2,72043746	21:00
6876	7257	-381	5,54101222	22:00

6424	6824	-400	6,22665006	23:00
Tuesday 18/8/2026				
5688	6400	-712	12,5175809	0:00
5366	5441	-75	1,39768915	1:00
5125	5293	-168	3,27804878	2:00
4975	5160	-185	3,71859296	3:00
4859	5054	-195	4,01317143	4:00
4887	4942	-55	1,12543483	5:00
5033	4917	116	2,3047884	6:00
5069	4977	92	1,81495364	7:00
4744	4949	-205	4,32124789	8:00
4093	4612	-519	12,6801857	9:00
3464	3946	-482	13,9145497	10:00
3049	3307	-258	8,46179075	11:00
2855	2943	-88	3,08231173	12:00
2964	2867	97	3,27260459	13:00
3196	3148	48	1,50187735	14:00
3426	3622	-196	5,72095738	15:00
4038	4145	-107	2,64982665	16:00
4962	5017	-55	1,10842402	17:00
6089	6028	61	1,00180654	18:00
6924	6933	-9	0,12998267	19:00
7270	7391	-121	1,66437414	20:00
7189	7438	-249	3,46362498	21:00
6720	7199	-479	7,12797619	22:00
6286	6742	-456	7,25421572	23:00
Monday 17/8/2026				
5512	6319	-807	14,6407837	0:00
5220	5299	-79	1,51340996	1:00
4982	5168	-186	3,73344039	2:00
4866	5036	-170	3,49362926	3:00
4777	4957	-180	3,76805526	4:00
4803	4869	-66	1,37414116	5:00
4934	4847	87	1,76327523	6:00
4971	4899	72	1,44840072	7:00
4543	4877	-334	7,35197006	8:00
3820	4447	-627	16,4136126	9:00
3164	3705	-541	17,0986094	10:00
2725	3044	-319	11,706422	11:00
2619	2669	-50	1,90912562	12:00
2729	2675	54	1,97874679	13:00
3021	2950	71	2,35021516	14:00
3468	3474	-6	0,17301038	15:00
4081	4205	-124	3,03847096	16:00
4925	5076	-151	3,06598985	17:00
5959	6015	-56	0,93975499	18:00
6685	6859	-174	2,60284218	19:00
7021	7261	-240	3,41831648	20:00
6909	7309	-400	5,78954986	21:00
6475	7051	-576	8,8957529	22:00
6053	6608	-555	9,1690071	23:00

Sunday 16/8/2026				
5512	6190	-678	12,3004354	0:00
5220	5299	-79	1,51340996	1:00
4982	5168	-186	3,73344039	2:00
4866	5036	-170	3,49362926	3:00
4777	4957	-180	3,76805526	4:00
4803	4869	-66	1,37414116	5:00
4934	4847	87	1,76327523	6:00
4971	4899	72	1,44840072	7:00
4543	4877	-334	7,35197006	8:00
3820	4447	-627	16,4136126	9:00
3164	3705	-541	17,0986094	10:00
2725	3044	-319	11,706422	11:00
2619	2669	-50	1,90912562	12:00
2729	2675	54	1,97874679	13:00
3021	2950	71	2,35021516	14:00
3468	3474	-6	0,17301038	15:00
4081	4205	-124	3,03847096	16:00
4925	5076	-151	3,06598985	17:00
5959	6015	-56	0,93975499	18:00
6685	6859	-174	2,60284218	19:00
7021	7261	-240	3,41831648	20:00
6909	7309	-400	5,78954986	21:00
6475	7051	-576	8,8957529	22:00
6053	6608	-555	9,1690071	23:00
Saturday 15/8/2026				
5512	6190	-678	12,3004354	0:00
5220	5299	-79	1,51340996	1:00
4982	5168	-186	3,73344039	2:00
4866	5036	-170	3,49362926	3:00
4777	4957	-180	3,76805526	4:00
4803	4869	-66	1,37414116	5:00
4934	4847	87	1,76327523	6:00
4971	4899	72	1,44840072	7:00
4543	4877	-334	7,35197006	8:00
3820	4447	-627	16,4136126	9:00
3164	3705	-541	17,0986094	10:00
2725	3044	-319	11,706422	11:00
2619	2669	-50	1,90912562	12:00
2729	2675	54	1,97874679	13:00
3021	2950	71	2,35021516	14:00
3468	3474	-6	0,17301038	15:00
4081	4205	-124	3,03847096	16:00
4925	5076	-151	3,06598985	17:00
5959	6015	-56	0,93975499	18:00
6685	6859	-174	2,60284218	19:00
7021	7261	-240	3,41831648	20:00
6909	7309	-400	5,78954986	21:00
6475	7051	-576	8,8957529	22:00
6053	6608	-555	9,1690071	23:00

Useful Graphs (MW / h):

